Geophys. J. R. astr. Soc. (1979) 58, 769–784

Resolution of superimposed magnetizations in the Devonian John O'Groats Sandstone, North Scotland

K. M. Storetvedt *Department of Geophysics, University of Western Ontario, London, Ontario N6A 5B7, Canada, and Department of Geophysics, University of Bergen, 5014 Bergen, Norway*

C. M. Carmichael *Department of Geophysics, University of Western Ontario, London, Ontario N6A 5B7, Canada*

Received 1979 February 1; in original form 1978 September 25

Summary. The John O'Groats Sandstone, on the north-east tip of Scotland, is the uppermost unit of a thick Old Red Sandstone sequence from the uppermost Middle Devonian. Samples from 13 sites have been treated by thermal and acid leaching techniques making use of the convergence of remagnetization circles when stable end points were not reached. In some cases, in which the magnetic constituents were asymmetrically distributed, splitting the specimens proved useful. Two axes of magnetization have been identified. The easiest to determine, called the A axis, is present most frequently in the reversed sense, at roughly Dec 190°, Inc −40°. The normal component can also be detected as stable end points on acid leaching. The A-axis magnetization is carried mainly by the pigmented cement. The B axis, which is aligned NNE–SSW and almost horizontal, is less well defined. Both the A- and B-axis orientations cited are after making corrections for structural tilts that are present at some of the sites. The direction of the B-axis magnetization agrees well with that of other Devonian units from west of the Great Glen Fault and is interpreted as having been acquired during, or shortly after, deposition. The A magnetization is interpreted as having been acquired during the Mesozoic, probably associated with regional burial and subsequent tectonic uplift of the area during the Jurassic. Later tectonic movements to produce at least the major structural tilts post-date the A magnetization and may be related to crustal unrest at the opening of the modern Atlantic. The John O'Groats directions are compatible with similar directions found in the Duncansby Neck, a volcanic vent that penetrates the Sandstone.

1 Introduction

The geology of Caithness (Fig. 1) is characterized by a thick succession of Old Red Sandstone sediments. At the base of the sequence there are barren red mudstones, arkoses and

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

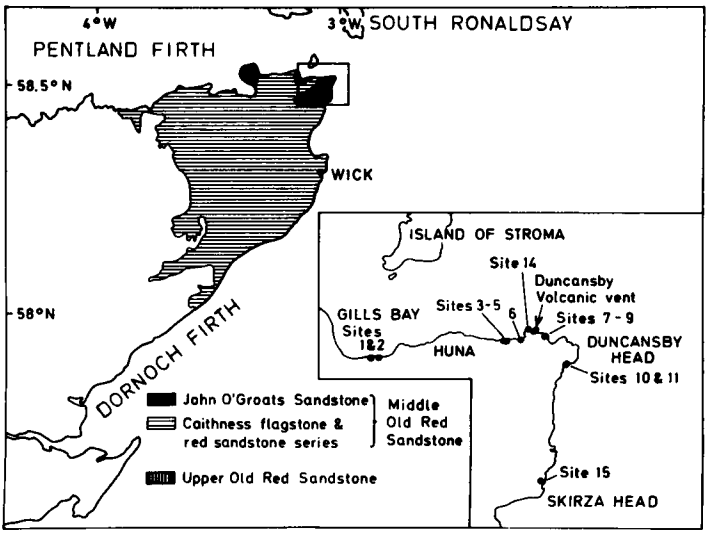


Figure 1. Location map of John O'Groats Sandstone including distribution of palaeomagnetic sampling sites.

conglomerates which Westoll (1951) believed to be of Lower Old Red Sandstone age. The overlying Caithness Flagstone constitutes the predominant lithological group and has a rich Middle Old Red Sandstone fish fauna (Crampton & Carruthers 1914; Jarvik 1948; Westoll 1951). In north-east Caithness the Flagstone group is overlain by the John O'Groats Sandstone whose age, based on fish remains in a few thin horizons, has been regarded as uppermost Middle Devonian (Crampton & Carruthers 1914; Phemister 1960; Donovan, Foster & Westoll 1974).

A palaeomagnetic reconnaissance study by Stubbs (1958) showed that the untreated NRM of the Caithness sediments, in spite of a considerable scatter, were basically reversely magnetized. Waage & Storetvedt (1973), who continued the palaeomagnetic study of Caithness sediments by applying thermal demagnetization, found that the red strata near the base of the sequence contained at least two superimposed magnetizations of opposite polarity. However, the blocking temperature spectra were such that single component remanences were extremely difficult to isolate. More recent investigations (Tarling *et al.* 1976; Turner, Vaughan & Tarling 1978) have emphasized the remagnetization aspects of these rocks. Unfortunately the lacustrine units studied by Turner *et al.* (1978) contained sulphides which decomposed on heating and prevented the use of the thermal demagnetization technique above *c.* 600°C, which is the temperature range in which the magnetic carrier of these rocks would most likely be haematite.

In the present paper we report on the experimental work that has been undertaken to determine the magnetization history of the John O'Groats Sandstone. This study is closely related to the work on the Duncansby Neck (*cf.* Fig. 1), a volcanic plug that penetrates the John O'Groats Sandstone (Storetvedt *et al.* 1978).

2 Sampling and laboratory procedures

The rock collection comprises block samples taken from 13 sites (numbered 1–11 and 14–15) along coastal sections between Gills Bay and Skirza Head (Fig. 1). The collected material consists dominantly of fine-grained red sandstones; exceptions are site 15 which

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

represents the yellow sandstone in the lower part of the sequence and sites 1–3 which are fairly calcareous. Further collection details are given in Table 1.

About 100 specimens in the form of cylinders 1.9 cm in both length and diameter have been subjected to stepwise thermal demagnetization in a furnace in which cooling to room temperature takes place in a field of less than 10 nT. About 40 specimens were chemically demagnetized in concentrated hydrochloric acid (10 N). In order to speed up the leaching process the cylinders were perforated by using a diamond drill-bit of 8 mm diameter and 1 mm wall thickness (Morris & Carmichael 1978). In conjunction with the leaching treatment many of the specimens were cut into halves, perpendicular to their axis, and studied separately. After each demagnetization step the remanence parameters of each specimen was measured using an astatic, a Digico or a Schonstedt DSM-1 magnetometer.

Table 1. Details of sites at which samples of John O'Groats Sandstone were collected.

Site no.	Sample nos	Strike and dip of strata	Locality
1	1–5	Horizontal	Gills Bay
2	6–10	Horizontal	Gills Bay
3	11–15	311°, 22° ENE	At John O'Groats pier
4	16–20	350°, 28° E	50 m east of site 3
5	21–25	300°, 16° NE	100 m east of site 3
6	26–30	290°, 16° NNE	~ Midway between Last House and Ness of Duncansby
7	31–35	263°, 47° N	Bay of Sannick
8	36–40	255°, 46° NNW	Bay of Sannick
9	41–45	253°, 22° NNW	Bay of Sannick
10	46–50	333°, 11° SW	Just south of Duncansby Head 200 m north of Thirle Door
11	50–54	337°, 6° SW	Just south of Duncansby Head 200 m north of Thirle Door
14	56–70	291°, 21° NNE	At Ness of Duncansby
15	71–72	Horizontal	Skirza, ~ 500 m south of Skippie Geo

3 Decomposition of the NRM

Sites 1–3 and 15, where the material is more calcareous or yellow in colour, have not given results that can be used for palaeomagnetic interpretation. These rocks either become too weakly magnetic after a few demagnetization steps or their magnetic moments changed erratically both on thermal and on leaching treatment.

Material from all the remaining sites behaves in a more systematic way; the specimens either show stable end-point magnetization or they exhibit more or less systematic vectoral swings as demagnetization progresses. Some of the swinging specimens move along well-defined great circle paths, suggesting that the remanence is dominantly a two-component system, but in other cases the directional variations are more complex. On the whole, chemical demagnetization, combined with specimen slicing, has been able to extract more detailed information from individual specimens than the thermal method. Chemically treated specimens could be measured reproducibly down to lower intensity levels than could thermally treated ones, the minimum intensities used are 1×10^{-4} and 5×10^{-4} A/m respectively. Because of difference in the extent of demagnetization obtained by the two methods Figs 2–5 show more results from the leaching procedure than from the thermal treatment. These figures show interesting examples of multi-component magnetization in several specimens. Only those which are regarded to have achieved a stable magnetization are included in Table 2.

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

Table 2. Palaeomagnetic results from the John O'Groats Sandstone giving characteristic directions (*D*, *I*) of individual specimens (see test), the corresponding temperatures (°C) or HCl leaching time (hr), and the palaeomagnetic group (*A* or *B*). All directions are after structural correction. See text for general details.

Site	Specimen	*D*	*I*	Group	Range (°C or hr of HCl)
1	JO 1-A3	191,	−44	A	NRM–500°C
2–3	Too weak to be tested				
4	JO 16-A1	202,	−31	A	NRM–164 hr
	JO 12-B2	204,	−42	A	NRM–620°C
	JO 17-A2	193,	−45	A	100–630°C
	JO 18-A2	199,	−41	A	100–600°C
	JO 18-B2	199,	−40	A	100–600°C
	JO 19-A3	203,	−41	A	100–600°C
	JO 19-B2	202,	−40	A	100–600°C
	JO 20-A1	208,	−38	A	100–600°C
	JO 20-A2	209,	−38	A	100–600°C
5	JO 21-A3	194,	−31	A	100–500°C
	JO 22-A1	208,	+4	B	152–196 hr
	JO 22-C1	203,	−2	B	112–152 hr
	JO 22-C2	199,	−36	A	100–500°C
	JO 23-B2	184,	−31	A	100–600°C
6	JO 27-B1	217,	+52	?	NRM–164 hr
7	JO 31-A1	210,	−33	A	NRM–98 hr
	JO 32-B3	190,	−26	A	NRM–620°C
	JO 32-C1	205,	−41	A	100–600°C
	JO 32-C3	207,	−38	A	100–600°C
	JO 33-B1	176,	−1	B?	400–620°C
8	JO 36-D1	195,	−40	A	100–620°C
	JO 36-E1	185,	−41	A	100–550°C
	JO 38-C1	188,	−41	A	100–600°C
	JO 39-A1	180,	−39	A	100–600°C
	JO 39-B3	187,	−39	A	100–620°C
	JO 40-A1	234,	+12	B	47–276 hr
	JO 40-C1	186,	−32	A	79–260 hr
9	JO 41-C2	(213,	+4)	B	44, 408 hr
	JO 42-A1	349,	+49	A	210–339 hr
	JO 42-B1	357,	+46	A	257–550 hr
	JO 44-A2	226,	+17	B	500–620°C
	JO 45-A1	151,	+68	?	NRM–164 hr
	JO 45-A2	181,	+12	B?	500–650°C
	JO 45-B1y	343,	+19	A?	358–510 hr
	JO 45-C2y	007,	+52	A	305–685 hr
10	JO 46-A1y	217,	+14	B	227–417 hr
	JO 45-A1x	302,	−65	?	227–728 hr
	JO 46-C2	177,	−55	A	100–600°C
	JO 47-E	171,	−48	A	100–500°C
	JO 47-B1	175,	−46	A	100–600°C
	JO 48-B1	162,	−41	A	100–600°C
	JO 48-A3	164,	−44	A	100–600°C
	JO 49-A1	173,	−52	A	100–600°C
	JO 49-B1	170,	−50	A	100–600°C
	JO 49-B2	197,	−33	A	NRM–79 hr
	JO 50-A1	169,	−41	A	100–600°C
	JO 50-B1	173,	−41	A	100–600°C
11	JO 52-B1	052,	+82	?	100–500°C
	JO 54-A1	163,	−42	A	100–600°C
	JO 54-C1	183,	−38	A	NRM–260 hr
	JO 55-C2	332,	+70	?	NRM–98 hr

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

Table 2 – *continued*

12–13	Not belonging to formation			
14	JO 66-C1	202, +17	B	140, 165, 428, 519 hr
	JO 67-A1	212, −5	B	NRM−47 hr
	JO 68-B1	220, −43	A	NRM−250 hr
	JO 69-A1	(023, +67)	A?	344, 358 hr
	JO 69-B1	(354, +29)	A	212, 226 hr
	JO 69-D1	215, +3	B	154, 344, 358 hr
	JO 70-A2y	(019, +24)	A	196, 344 hr
	JO 70-A2x	(223, −11)	B	196, 344 hr

Note:

The five samples, for which only a question mark is given in the Group column, were not used in calculating the *A* pole in Table 3 (see text).

The most important magnetization axis in the John O'Groats Sandstone is directed roughly north–south with an intermediately steep inclination. This axis of magnetization will be referred to as the *A* axis. The reversed *A* magnetization (*c*. 190, −40) is the predominant one. The remanence direction in many specimens remain stationary in this position over the major range of demagnetization, particularly on thermal treatment. It is frequently found, however, that when demagnetizing the last 10 to 15 per cent of the magnetic moment the remanence vector moves along a great circle path (*cf.* JO 25-A3 of Fig. 3(b)). There are only a few specimens in this study for which the normal *A* magnetization is defined by an end-point on demagnetization and only the chemical method has been successful in isolating these end points. Specimen JO 42-A1 (Fig. 2) is an example of a specimen that reaches the normal *A* component. It is a general observation in this study that the red pigment disappears relatively quickly and even in the most resistant specimens it is completely removed in less than 200 hr of leaching. Also, the remanence intensity decays rapidly in the initial stages of HCl treatment. Removal of the pigment is in general connected with a moving away of the remanence vector from the reversed *A*-axis position. It is concluded, therefore, that the reversed *A* magnetization must be carried by the red iron-oxide pigment while the normal *A*-axis component is associated with larger magnetic grains.

Fig. 3 illustrates two cases, JO 18-A1 and JO 41-C1, where specimen slicing seems to provide very useful information on within-specimen magnetization. JO 18-A1 starts out in the reversed *A*-axis position and moves slowly but perfectly along a great circle path. After 158 hr of leaching the specimen was cut. On continued demagnetization the two smaller specimens define remagnetization trends that are differently oriented compared to that of the combined specimen and the three extended remagnetization circles intersect in the position of the normal *A*-axis component. In palaeomagnetic studies one often encounters specimens like JO 41-C1 (Fig. 3(b)) which on demagnetization remain in a discordant position. After 480 hr of leaching, which had reduced the remanence intensity by 85 per cent, the specimen was cut. The two halves (x and y) define separate stable directions which, together with the pre-cut magnetization, are distributed along a great circle that passes through the *A*-axis components. In this case leaching removes the components at an equal rate but fortunately the two components are not homogeneously distributed throughout the volume of the specimen so that the composite nature of its remanence can be detected by specimen slicing.

The second axis of magnetization in this formation, the *B* axis, is nearly horizontal and strikes NNE–SSW. This magnetization is much less distinct than the *A* magnetization and for its detection chemical treatment is a much more powerful technique than thermal

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

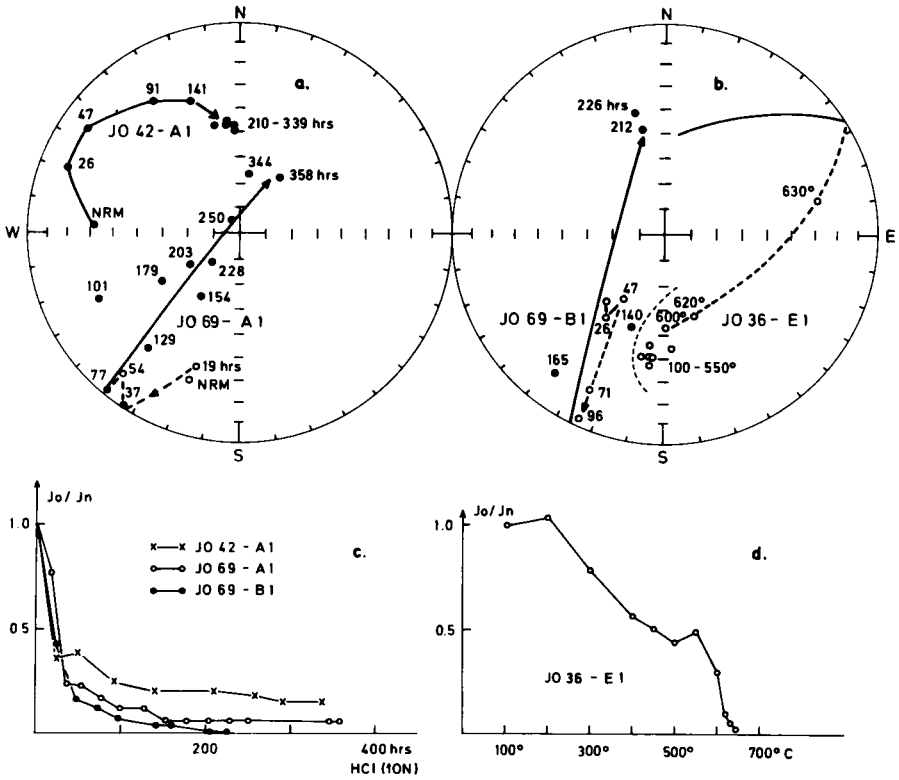


Figure 2. Demagnetization data for examples of specimens that define the *A* axis of magnetization. It is defined as the normal component by end points of acid leaching treatment (specimens JO 42-A1, JO 69-A1, JO 69-B1) and also as the reversed component over a number of the initial treatment steps by acid leaching (specimen JO 69-B1) and by thermal treatment (specimen JO 36-E1). In addition the last steps of the thermal treatment of JO 36-E1 define a great circle heading toward the normal *A*-axis position. Parts a and b of the figure are equal area plots in which open (closed) symbols indicate upward (downward) pointing magnetization vectors. Numbers beside points indicate hours of acid leaching if they have no designation other than hours after the highest number, and indicate the temperature of the thermal demagnetization step in °C if the number has a ° symbol. Parts c and d show the relative intensity of the magnetization of the specimens in parts a and b after each stage of acid leaching and thermal treatment respectively. These specimens are included in Table 2.

demagnetization. Figs 4 and 5 illustrate some examples in which the *B* magnetization is involved. Both specimen JO 41-C2 and JO 66-C1 (Fig. 4(a), (b)) give some indication of possessing the two anti-parallel *B* vectors. The directional change on demagnetization for JO 70-A2 (Fig. 4(c)) follows a northerly trend but after 196 hr the intensity was below the level accepted for reliable measurement (1×10^{-4} A/m). At this stage the specimen was cut: the upper half was found to be reversely magnetized in good agreement with the *B* magnetization while the lower half was normally magnetized closer to the *A* axis. Further leaching up to 344 hr did not change these directions significantly. It is important to recognize that the split of the JO 70-A2 magnetization into opposite polarity directions is associated with intensity increases (see Fig. 4(e)). There is of course no guarantee that the specimen split provide clearly separated sub-components so JO 70-A2 may either contain the two *B*-axis components or, alternatively, the normal *A* vector plus the reversed *B* magnetization. Specimen JO 46-A1 (Fig. 5(b)) has also an uneven distribution of its magnetizations. Thus the upper half, which carries the bulk of the intensity, is discordantly

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

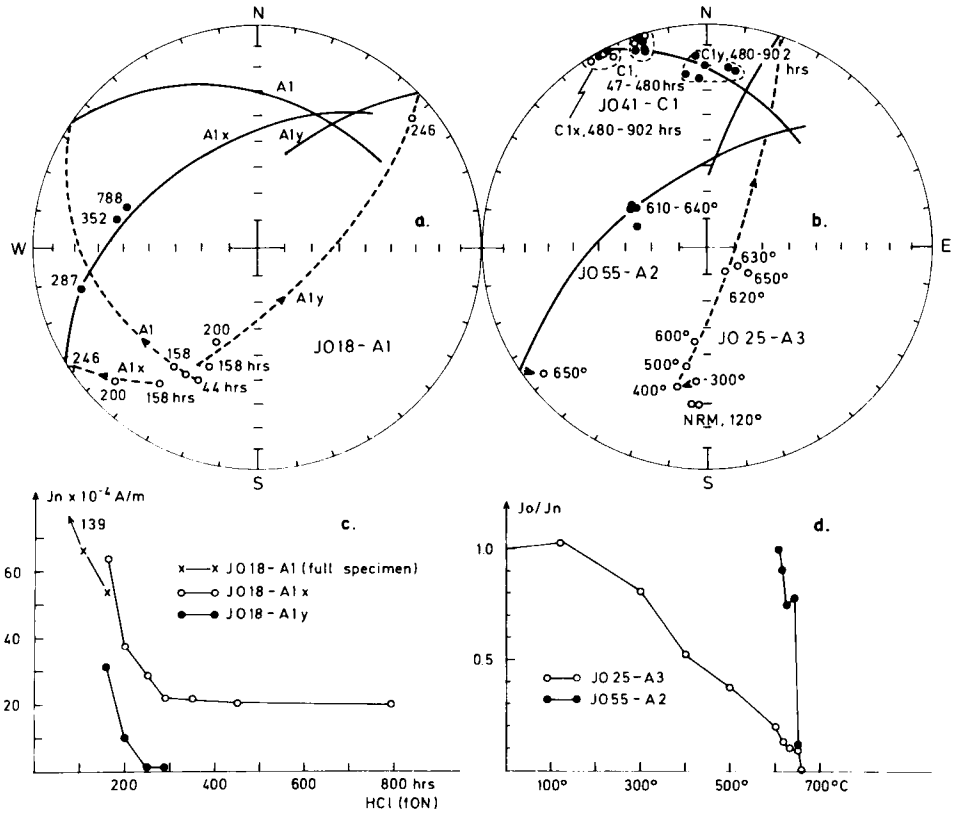


Figure 3. Demagnetization data for specimens that indicate the A axis by defining great circles. The specimen cutting technique was used for specimens JO 18-A1 and JO 41-C1 (see text). Diagram conventions are the same as in Fig. 2. These specimens are not included in Table 2 but those that meet Halls (1976, 1978) quality factor of $Q < 1 \times 10^{-5}$ are used in the great circle analysis that yields the A^* pole in Table 3.

magnetized (probably multivectorial) while the much weaker lower half is magnetized in the reversed B direction. Specimen JO 39-B2 appears to have all four A and B vectors superimposed; below 337 hr of leaching the demagnetization pattern seems to be dominated by the interplay of the two A components but on prolonged chemical treatment the B components tend to determine the orientation of the remagnetization trends.

4 Palaeomagnetic results

Sixty specimens tend to give stable magnetic directions. Of these about 70 per cent had stable magnetizations throughout the complete demagnetization range, or reached end points upon thermal and/or chemical demagnetization. For the latter group the terminal directions were mostly defined by the mean of three or more demagnetization steps. Exceptions are a few specimens, the results of which are given in brackets in Table 2, that seem to define characteristic directions of magnetization for the formation but where the 'stable' end points are based on two steps only (on continued demagnetization the moment became too low for reliable measurement). The remaining 30 per cent had stable directions of magnetization until less than 10 per cent of their original intensity remained and then their directions changed in an erratic and non-reproducible manner. For these the stable

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

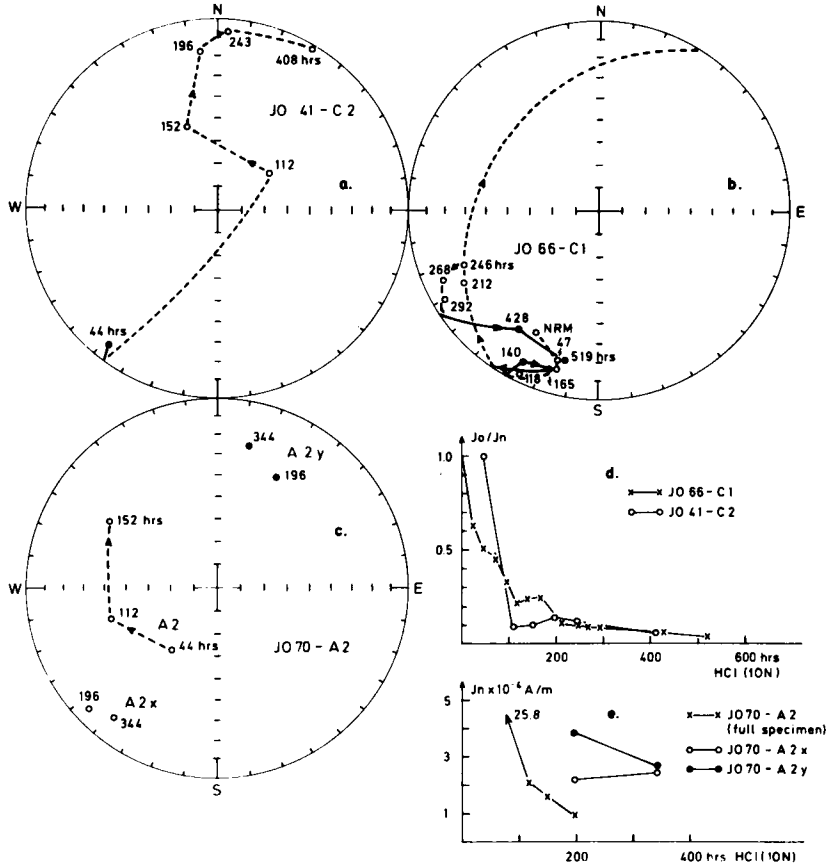


Figure 4. Demagnetization data for examples of specimens that define the *B* axis of magnetization by different procedures involving acid leaching. Specimen JO 41-C2, shown in (a), moves from the SSW *B*-axis position to an antiparallel point in the NNE position; specimen JO 66-C1, shown in (b), shows the interplay of the two polarities of the *B* axis and probably some of the reversed *A* component. (d) and (e) show the intensity of magnetization of the specimens at each stage of the leaching process. Note that upon cutting specimen JO 70-A2 the intensity of the full specimen is the vector sum of the larger but different polarity magnetizations of the two parts. Diagram conventions are the same as in Fig. 2.

magnetization associated with the initial 90 per cent of the NRM intensity was used. The data for the 60 specimens are listed in Table 2. There were 31 additional specimens whose directions moved along paths directed generally to the north but which did not reach stable end points before their intensity fell below the level of reproducible measurement. The data for these specimens are not included in Table 2 but were used for the great circle analyses (see below) which yields the poles given as A^* and B^* in Table 3. Attempts at vector subtraction for this latter group of specimens did not prove successful in finding the reversed sub-components which suggests that their component magnetizations have too much overlap of their stability spectra.

The sedimentary strata at the different collecting sites dip at various angles up to about 45° (see Table 1). To determine the timing of this tectonic tilting relative to the time of magnetization, the directions of all those specimens from Table 2 having intermediate to steep inclinations, i.e. those interpreted as defining the *A* axis, were plotted before and after tectonic corrections. This plot is shown in Fig. 6 in which plot a is before, and plot b is after,

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

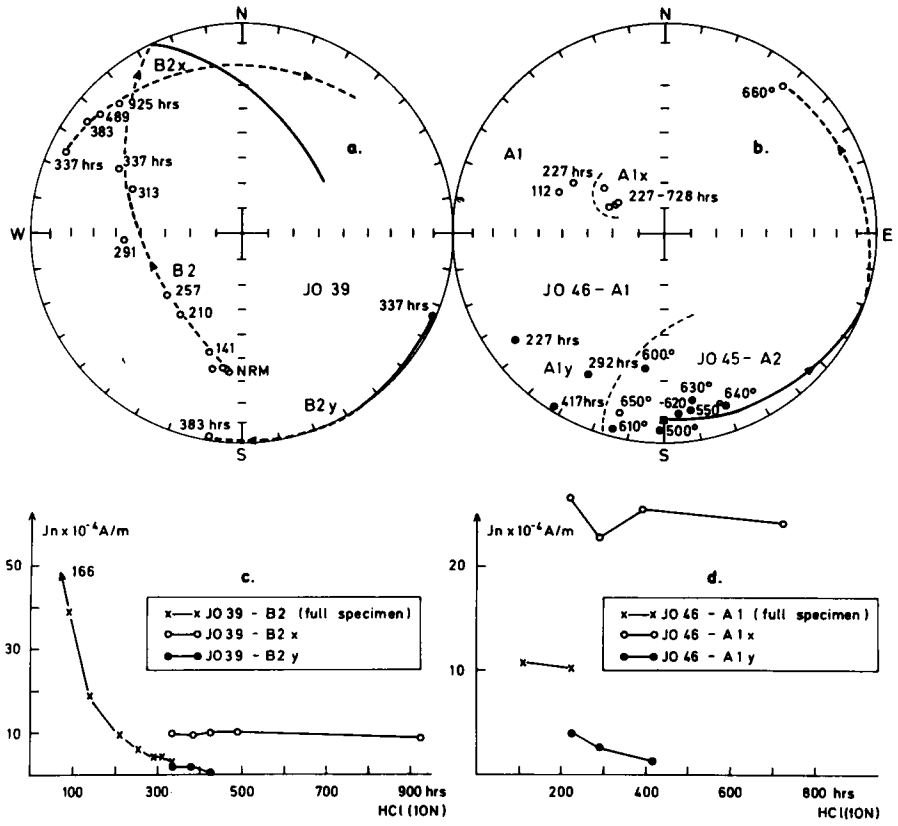


Figure 5. Demagnetization data for examples of specimens in which the techniques of acid leaching, thermal treatment and specimen cutting only partly resolve the multicomponent magnetizations that define both polarities of the *A* and *B* axes of magnetization (see text). Diagram conventions are the same as in Fig. 2.

tectonic corrections. Making the correction produces a marked improvement toward a two-polarity *A*-axis grouping. There are, however, five specimens, indicated by a question mark in the column headed Group in Table 2, and by an ∇ over their symbol in Fig. 6(b), which do not fall into the grouping. This may be due to some part of their magnetization being acquired after tilting or to unresolved components of magnetization. Whatever the cause of their anomalous directions of magnetization these specimens have been excluded in the calculation of the *A*-axis pole position given in Table 3. The reduced scatter of the grouping by the tectonic correction indicates that at least the major part of the structural dip of the John O'Groats Sandstone post-dates the *A* magnetization. Since the *A* component is interpreted as being the younger of the two characteristic magnetizations (see below), all the results of this study have been subjected to tectonic correction.

The reversed group of Fig. 6(b) has a certain elongation that may be caused by improper structural correction; for example that a minor part of the tilting may pre-date the *A* magnetization or alternatively the unfolding to the palaeo-horizontal may be more complex than simple rotation of bedding about present strike direction. In Fig. 7 the magnetization directions of the specimens are plotted as though they all had reverse polarity to separate them on the diagram from the location of the intersection points of northward trending remagnetization circles. The directional trends plotted here are only those which lie along

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

Table 3. Mean palaeomagnetic data from the John O'Groats Sandstone.

Group	N	D_m	I_m	κ	α_{95}	Pole	
A	43	188.3	−41.1	32	3.9	163.6 E, 54.2 N	Data from Table 2. Normal directions have been inverted
A^*	21	020.4	+ 38.6				Best intersection point of normally trending remagnetization circles
$B1$	12	209.2	+ 05.7	18.2	10.4	144.5 E, 24.1 N	Data from Table 2
$B2$	10	215.2	+ 05.3	34.3	8.3	138.2 E, 22.5 N	Data from Table 2 less JO 33-B1 and JO 45-A2
B^*	10	041.9	−06.0				Best intersection point of normally trending remagnetization circles

Note: No significance is attached to the polarity of the A^* and B^* groups.

great circle paths defined by at least three successive demagnetization steps, and which have a quality factor $Q < 1 \times 10^{-5}$ (Halls 1978). Thirty-one demagnetization trends satisfy these criteria. For some of the paths it was difficult to decide which group (A or B) they were heading toward. Such data have been assigned to the A group as the A magnetization is the predominant one of the formation. The only exception is the specimen with the steepest path which was left in the B group because the bulk of its remanence is directed in the reversed B position. It is important to realize that the remagnetization circles only define the axis and say nothing about the normal component (there are two antiparallel and,

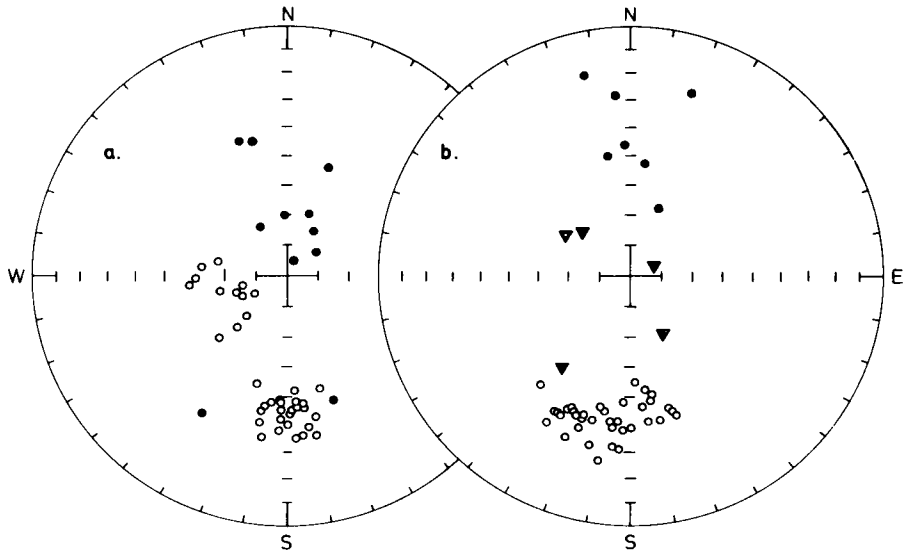


Figure 6. Equal area plots of the magnetizations of all the samples listed in Table 2 that have been identified as A axis, i.e. they have intermediate to steep inclinations. In (a) no tectonic correction has been made and in (b) the tectonic correction indicated by the dip and strike of the strata at the collection sites, as listed in Table 1, has been applied. Open (closed) symbols indicate upward (downward) vectors. The five symbols in (b) that are indicated by an overprinted triangle are for specimens that become (or remain) discordant upon tectonic correction. These have a question mark in the Group column of Table 2 and are interpreted as specimens in which multicomponent vectors have not been resolved or some of their magnetization post dates the tilting. They have been excluded in the calculation of the A pole given in Table 3.

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

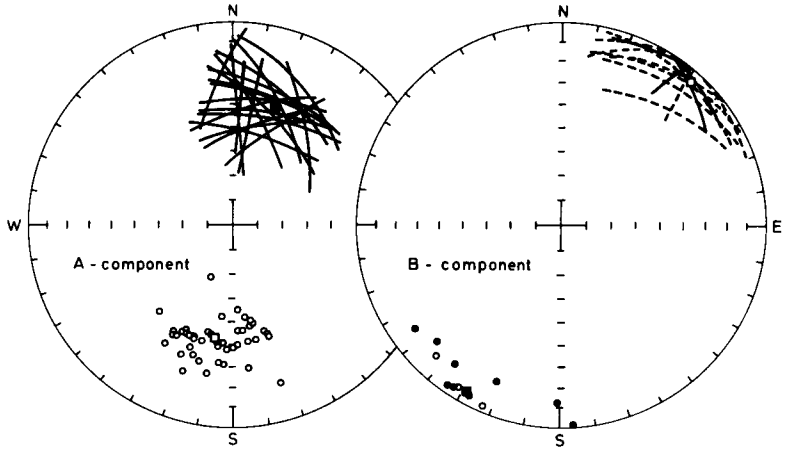


Figure 7. Equal area plots of the magnetizations of specimens listed in Table 2 in which all are plotted as southerly vectors with open (closed) symbols indicating upward (downward) vectors. Segments of the great circle paths defined by successive demagnetization steps for specimens used to calculate the A^* and B^* poles of Table 3 are indicated in the region of their northerly located intersection points. The specimens inferred as defining the A axis of magnetization are shown on the left and those that define the B axis on the right. In each case the location of the mean vector is shown by a squared symbol.

equally valid, intersection points). The sense is only defined if there is a clearly less-dispersed component and that is not the case here.

Overall palaeomagnetic results are listed in Table 3. There is very good agreement between the mean direction of the reversed A group (Table 2 data) and the corresponding best intersection point of the remagnetization trends. The B magnetization is of subordinate importance compared to the A magnetization. Only 12 specimen directions define this axis. The normal intersection point of directional trends agrees well with the characteristic directions derived from an independent sample population although this convergence point is much more ill-defined than that of the A axis. The angular difference between the mean A and B magnetizations is $c.50^\circ$ and the mean directions are significantly different at the 95 per cent probability level.

5 Magnetic mineralogy

The magnetic mineralogy of the John O'Groats Sandstone samples is quite different from that of the lacustrine units further down in the Caithness Old Red Sandstone section that were investigated by Turner *et al.* (1978). These authors found magnetite and relatively large amounts of iron sulphide that oxidized upon heating to produce a magnetically very soft component which was probably additional magnetite. This soft component picked up spurious magnetizations in the laboratory and prevented thermal demagnetization at temperatures that would have determined any natural remanence carried by haematite.

In contrast to the Turner *et al.* (1978) lacustrine units, the natural remanence of the John O'Groats samples is carried exclusively in various haematite phases. These can be divided into two categories, haematite in relatively large sedimentary grains and haematite in the cementing matrix.

Considering the large-grain category first, these are interpreted optically as being originally iron titanium oxides in both a spinel form, i.e. titanomagnetite, and a rhombohedral form, i.e. ilmenite. Both of these forms have been severely oxidized to haematite and rutile. Grains that are inferred to have been titanomagnetite originally are roughly

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

equidimensional in shape and have the crystal axes of the haematite phases aligned along 111 directions. None of the original magnetite has survived. An example of this type of grain is shown in Plate 1(b). Grains inferred to have been ilmenite originally are elongated in shape and have separated into irregular areas of variable reflectivity without any particular alignment of the crystal axes of these areas. Otherwise they are similar in appearance to Plate 1(b). In all of the samples a proportion of the large grains has been completely converted to rutile. This proportion varies in different samples from around 40 to 100 per cent. To convert the grains to rutile would require not only oxidation but also some process, such as attack by acidic solutions, to remove the iron. Attack by acidic solutions is suggested because the large grains that are now rutile in the natural samples are indistinguishable from all the large grains in samples that have been completely leached in hydrochloric acid. An example of a large grain that is mostly rutile but still has some haematite at the centre is shown in Plate 1(c) and (d). The timing of the alteration of the large grains would be after deposition, because the rutile end product is no longer a coherent grain and could not be deposited in that form, but before the red cement was introduced because haematitic staining of the cement would be much more vulnerable to the acid solution than the larger grains would be.

The haematite associated with the cementing matrix is present in several forms. Particle sizes of haematite large enough to reflect light have developed throughout most of the cement as scattered dust and also as veins and rims of some sediment particles (Plate 1(e)). All of the haematite associated with the cement is in small enough particles to show ruby red translucence when viewed with crossed polarizers. Included in the haematite associated with the red cement is some that has formed as an alteration product of sedimentary grains. These seem to have been altered during the cementing process since they are now only relict structures within the cementing matrix (Plate 1(f)). The haematitic staining of the cement may not contribute to the magnetization. This conclusion comes from examination of those samples that had initial moments of the order of 1×10^{-3} A/m and thus were too weak to apply the demagnetization techniques even though they have abundant red staining of the cement (see C of Plate 1(a)). These were samples from sites 2 and 3 of Table 1. In all of them the original large grains had been converted to rutile and there are very few haematite particles large enough to reflect light. Microscopic examination of polished thin sections of samples at various stages of the acid leaching process show clearly that the haematite associated with the cementing matrix is destroyed more rapidly than are the larger grains. Since the acid attacks the larger grains within volumes of the sample from which the haematite in the cement has been removed, however, part of the magnetization carried by the large grains will be destroyed before all of that carried by the cement is removed. The technique of making drill cuts in the samples, which is illustrated in Plate 1(a), minimizes this overlapping in the demagnetization process by allowing rapid penetration of the acid to the interior of the sample. Because of the observed order in which the haematite is removed in the leaching process it is concluded that the more resistant B-axis magnetization is carried by some of the larger grains. The A axis is inferred to be carried by the haematite associated with cementing matrix as well as by some of the larger grains that probably suffered further oxidation during the cementing process. This association of the two axes of magnetization with the different categories of haematite is supported by examination of sample JO 41 in which the B axis predominates (Fig. 4(a)). It contains many large grains in which there is little or no alteration to rutile and the red cement contains less polishable haematite.

Thermal demagnetization curves, such as those shown in Figs 2(d) and 3(d), show the presence of components that unblock over temperature ranges usually found for magnetite. There is no evidence of magnetite being present, however, because none is observed in

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

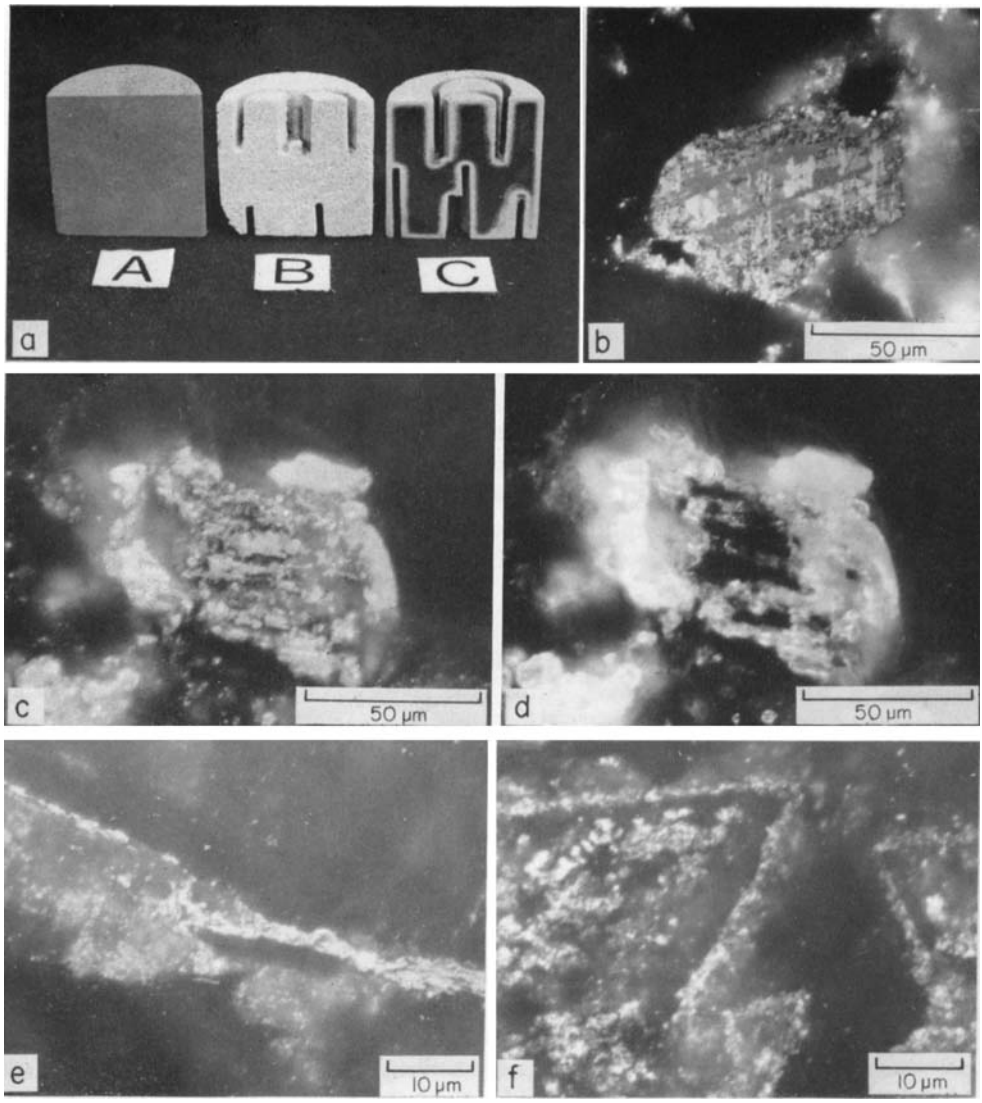


Plate 1. John O'Groats Sandstone: (a) specimens cut vertically to show their interior, specimens A and B show, respectively, the dark (red) colour of the natural sandstone, and the very light colour after the acid leaching process is complete. Specimen B is number JO 41-C2 whose magnetization changes during leaching as shown in Fig. 4 (a) and (d). Specimen C is number JO 4-B1, a fine grained example of the material from Site 1 of Table 1 that is very weakly magnetized even though it contains abundant red pigment in the cement. This photograph (specimen C) was taken after 44 hr of acid leaching and it shows the effective way the drill cut technique (Morris & Carmichael 1978) allows the acid to penetrate (see text). Photomicrographs of polished thin sections show in (b) a sedimentary grain of haematite with some rutile on the edges. It is an example of a grain inferred to have been a titanomagnetite because of the 111 alignment of the haematite regions that can be distinguished here under partially polarized light. In (c) and (d) are shown a grain from a natural sample of JO 54. All but the small regions of haematite near the centre, that show black with crossed polarizers in part (d), has been converted to rutile, which is brighter than the haematite in reflected light in (c) and very bright due to internal reflections, that make it translucent under crossed polarizers, in (d). This grain, although in a natural sample, is indistinguishable from grains that have had almost all the haematite removed in the acid leaching process (see text). Parts (e) and (f) show as points of light or as reflecting areas, the grains of haematite in the cementing matrix. A vein of haematite formed in a narrow space between sedimentary grains is shown in (e) while in (f) the haematite particles seem to have formed as an alteration product of crystals within the matrix. These are now only relict structures outlined by the haematite decoration.

[*facing page* 780]

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

polished section and negligible moment is induced in the samples in magnetic fields up to 800 mT. These lower temperature unblocking components are interpreted as being a combination of the effect of small particle size among the haematite grains in the cement and titanium in the haematite formed in the altered large grains. The relatively high degree of optical anisotropy of the haematite in grains such as the one shown in Plate 1(b) suggest considerable Ti content. Such causes for the distribution of unblocking temperatures would account for the general poorer success in separating the *A* and *B* components by the thermal demagnetization technique as compared to the acid leaching procedure.

6 Magnetization history

The palaeomagnetic results outlined above agree very well with recent palaeomagnetic findings in the Duncansby volcanic vent (Storetvedt *et al.* 1978). This small basic intrusion which cuts through the John O'Groats Sandstone is shown to possess a two-axis magnetization: the palaeomagnetically oldest axis corresponds extremely well with Old Red Sandstone data from Caithness and Orkney (Storetvedt & Petersen 1972; Waage & Storetvedt 1973) while the second axis seems related to a Mesozoic overprinting event (*cf.* Fig. 8). It has been concluded therefore (Storetvedt *et al.* 1978) that the Duncansby Neck most likely relates to the volcanism at the base of the Upper Old Red Sandstone in the nearby Island of Hoy (Halliday, McAlpine & Mitchell 1977).

As seen from Fig. 8 the John O'Groats *B* pole plots in agreement with other Old Red Sandstone poles from west of the Great Glen Fault. These poles have about a 15° longitudinal displacement to the left of the corresponding position for stable continental Europe. The John O'Groats results are thus compatible with the proposition that there is a discordance in Old Red Sandstone palaeomagnetism across the Great Glen Fault which has been associated with a major sinistral trans-current movement along this dislocation (Storetvedt 1974, 1975).

The *A* pole plots in a Mesozoic position relative to the European polar wander curve. A reliable assessment of the age of this magnetization overprint is hampered by an incomplete knowledge of Mesozoic palaeomagnetism for Europe. However, carefully tested data from suggested late Jurassic–Cretaceous volcanics from Spitsbergen (poles 10 and 11 of Fig. 8) suggest a marked eastward polar migration for Europe that is fully compatible with corresponding observations from North America. It is thought, therefore, that the Mesozoic polar curve suggested in Fig. 8 is essentially correct. On these assumptions the John O'Groats *A* pole can probably be assigned an age of around Lower-Middle Jurassic.

The strong Mesozoic magnetization overprint appears to be a general phenomenon in Old Red Sandstone rocks from Caithness (Storetvedt *et al.* 1978; Turner *et al.* 1978). Following the concept outlined in Storetvedt *et al.* (1978), in which this problem has been discussed at some length, it is suggested that the coastal districts of north-east Scotland formed part of the North Sea depositional basin in late Palaeozoic and early Mesozoic times. This provided a lengthy burial history for the Old Red Sandstone strata giving rise to a temperature elevation that enhanced chemical processes as well as increasing the rate at which viscous magnetization would be acquired. The secondary magnetization of the Duncansby Neck (Storetvedt *et al.* 1978) was interpreted as having been frozen in as a result of uplift during a period of regional tectonic unrest in the Lower-Middle Jurassic. As a result of this tectonic activity along the Great Glen Fault and in the Moray Firth basin (Bacon & Chesher 1975) the Jurassic sediments are down-faulted relative to the Old Red Sandstone sediments exposed on the landward side. Similar features are also exposed onshore along the Helmsdale Fault.

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

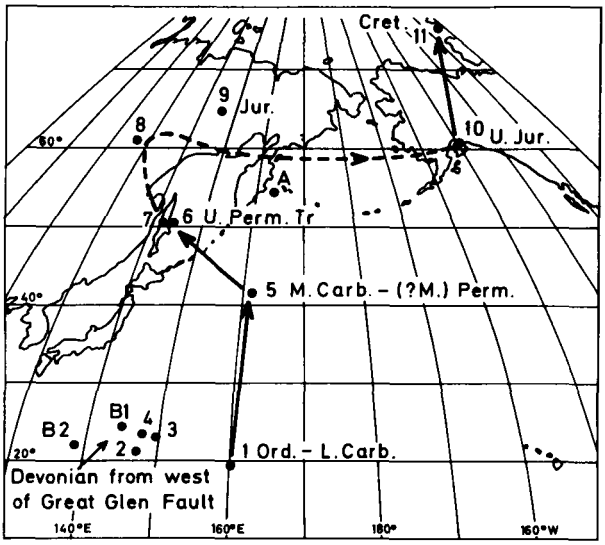


Figure 8. The John O'Groats poles A and B seen in conjunction with the suggested European polar wander path for part of Phanerozoic time. Poles 1, 5, 6 and 9 are mean poles from the following sources: 1 = Storetvedt (1967); 5, 6 and 9 = Phillips & Forsyth (1972) (other sources give similar results). Poles from individual formations are as follows: 2 = Waage & Storetvedt (1973); 3 = Storetvedt & Petersen (1972); 7 = Zijderveld (1975); 10 = Briseid & Halvorsen (1974); 11 = Sandal & Halvorsen (1973). Poles 4 and 8 are for the Duncansby Neck which intrudes the John O'Groats Sandstone (Storetvedt *et al.* 1978). Poles A, B_1 and B_2 are for the John O'Groats Sandstone and are taken from Table 3.

The Mesozoic magnetization overprint in the Duncansby Neck (pole 8 of Fig. 8) has a more westerly, i.e. probably older, pole position than that of the corresponding John O'Groats magnetization. This may reflect differences in the mode of acquisition of the secondary magnetizations. The Duncansby overprint is seen as a viscous remanence impressed during burial at elevated temperatures. The John O'Groats overprint is associated with fine haematite grains in the red cementing matrix. This material has unblocking temperatures ranging well above 600°C and clearly in excess of those associated with the demagnetization of the Duncansby Neck. It is unlikely, however, that the magnetization of the red cement has been acquired during cooling from such high temperatures: it is more probable that it is a CRM acquired as the haematite formed in the cement by a dehydration–oxidation process. If this process continued to temperatures lower than those required partly to remagnetize the Duncansby Neck, then the A magnetization of the John O'Groats Sandstone could have been acquired at a later state of the uplift process. This could explain the different pole positions for the secondary magnetizations of the two rock types.

The removal upon tilt correction of the majority of discordant, mostly steeply inclined, directions of magnetization suggest that at least the major tectonic phase, having affected the John O'Groats Sandstone, must be of post-Lower-Middle Jurassic age. This is likely to be connected with the crustal unrest that preceded the formation of the modern North Atlantic.

7 Conclusions

We conclude that the John O'Groats Sandstone acquired its oldest magnetization component during or shortly after deposition in Middle-Upper Devonian time. A subsequent large-scale transcurrent movement along the Great Glen Fault (probably in the uppermost

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

Devonian) led to a slight clockwise rotation of the north-west part of Scotland west of the Great Glen Fault. In subsequent times the coastal districts of north Scotland became part of the North Sea depositional basin and the Old Red Sandstone strata of Caithness, including the John O'Groats sequence, were buried under a pile of Upper Palaeozoic and Lower Mesozoic sediments of unknown thickness. This burial led to a rise in temperature resulting in an enhancement of thermo-chemical magnetization processes. Thus, at about the Triassic/Jurassic boundary the original palaeomagnetism had been nearly completely erased. A strong Mesozoic overprint was subsequently frozen in as a result of the uplift connected with the important volcano-tectonic phase that affected the northern North Sea in Lower-Middle Jurassic time. The acquisition of the secondary magnetization is therefore tied into the regional geological framework. At some time after uplift the John O'Groats Sandstone was affected by a structural deformation causing stratal dips ranging upwards to about 45°. This event is thought to be related to final continental breakup and initiation of sea-floor spreading in the North Atlantic.

Acknowledgments

This research was supported by the Norwegian Research Council for Science and the Humanities, Grant D.41, 12-6 and by the National Research Council of Canada Grant A1776. The field expenses were covered by K. Birkelands Fond. Technical assistance, at various stages of the work, by K. Breyholtz, F. Cleveland, L. E. Otterå and T. Storetvedt is gratefully appreciated. We are grateful to Dr P. A. Ziegler for discussions on the Great Glen Fault problem, and to Dr H. C. Halls and the reviewers for constructive comments on the manuscript.

References

Bacon, M. & Chesher, J., 1975. Results of recent geological and geophysical investigations in Moray Firth, Scotland, *Nor. geol. Unders.*, **316**, 99–104.

Briseid, E. & Halvorsen, E., 1974. The primary magnetic remanence of a dolerite sill from Northeast Spitsbergen, *Phys. Earth planet. Int.*, **9**, 45–50.

Crampton, C. B. & Carruthers, R. G., 1914. The geology of Caithness, *Mem. Geol. Surv. Scotland*, 194 pp.

Donovan, N., Foster, R. J. & Westoll, T. S., 1974. A stratigraphical revision of the Old Red Sandstone of north-eastern Caithness, *Trans. R. Soc. Edinb.*, **69**, 167–201.

Halliday, A. N., McAlpine, A. & Mitchell, J. G., 1977. The age of the Hoy lavas, Orkney, *Scott. J. Geol.*, **13**, 43–52.

Halls, H. C., 1976. A least-squares method to find a remanence direction from converging remagnetization circles, *Geophys. J. R. astr. Soc.*, **45**, 297–304.

Halls, H. C., 1978. The use of converging remagnetization circles in palaeomagnetism, *Phys. Earth planet. Int.*, **16**, 1–11.

Jarvik, E., 1948. On the morphology and taxonomy of the Middle Devonian osteolepid fishes of Scotland, *K. Svenska Vetskaps Akad. Handl.*, **25**, 1–301.

Morris, W. A. & Carmichael, C. M., 1978. Palaeomagnetism of some late Precambrian and lower Paleozoic sediments from L'Adrar de Mauritanie, West Africa, *Can. J. Earth Sci.*, **15**, 253–262.

Phemister, J., 1960. *British Regional Geology. Scotland: The Northern Highlands*, Her Majesty's Stationery Office, Edinburgh.

Phillips, J. D. & Forsyth, D., 1972. Plate tectonics, palaeomagnetism, and the opening of the Atlantic, *Bull. Geol. Soc. Am.*, **83**, 1579–1600.

Sandal, S. T. & Halvorsen, E., 1973. Late Mesozoic Palaeomagnetism from Spitsbergen; implications for continental drift in the Arctic, *Phys. Earth planet. Int.*, **7**, 125–132.

Storetvedt, K. M., 1974. A possible large-scale sinistral displacement along the Great Glen Fault in Scotland, *Geol. Mag.*, **111**, 23–30.

Storetvedt, K. M., 1975. Possible large-scale sinistral displacement along the Great Glen Fault in Scotland (reply to W. Mykura), *Geol. Mag.*, **112**, 91–94.

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020

Storetvedt, K. M., Carmichael, C. M., Hayatsu, A. & Palmer, H. C., 1978. Palaeomagnetism and K/Ar results from the Duncansby volcanic neck, NE, Scotland: superimposed magnetizations, age of igneous activity, and tectonic implications, *Phys. Earth planet. Int.*, **16**, 379–392.

Storetvedt, K. M. & Petersen, N., 1972. Palaeomagnetic properties of the Middle-Upper Devonian volcanics of the Orkney Islands, *Earth planet. Sci. Lett.*, **14**, 269–278.

Stubbs, P. M., 1958. *PhD thesis* (unpublished), University of London.

Tarling, D. H., Donovan, R. N., Abou-Deeb, J. M. & El-Batrouk, S. I., 1976. Palaeomagnetic dating of haematite genesis in Orcadian basin sediments, *Scott. J. Geol.*, **12**, 125–134.

Turner, P., Vaughan, D. J. & Tarling, D. H., 1978. Palaeomagnetic and mineralogical studies of Devonian lacustrine sediments from Caithness, Scotland, *Phys. Earth planet. Int.*, **16**, 73–83.

Waage, H. L. & Storetvedt, K. M., 1973. The Devonian geomagnetic axis relative to the Orcadian basin, N. Scotland, *Z. Geophys.*, **39**, 937–951.

Westoll, T. S., 1951. The vertebrate-bearing strata of Scotland, *Int. Geol. Congr., 1948*, **18**, 5–20.

Zijderveld, 1975. Paleomagnetism of the Esterel rocks. *Thesis*, University of Utrecht.

Downloaded from https://academic.oup.com/gji/article-abstract/58/3/769/677766 by Curtin University Library user on 03 August 2020